

5.2.2

For each positive integer n , let $P(n)$ be the formula

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

- a. Write $P(1)$. Is $P(1)$ true?
- b. Write $P(k)$.
- c. Write $P(k+1)$.
- d. In a proof by mathematical induction that the formula holds for every integer $n \geq 1$, what must be shown in the inductive step?

Let $P(n)$ be

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

a.

For the left hand side:

$$P(1) = \sum_{i=1}^1 i^2 = (1)^2 = 1^2 = \boxed{1}$$

For the right hand side:

$$P(1) = \frac{1(1+1)(2(1)+1)}{6} = \frac{2 \times 3}{6} = \frac{6}{6} = \boxed{1}$$

Solution: $P(1)$ is true.

b.

For the left hand side:

$$P(k) = \sum_{i=1}^k i^2 = \boxed{1^2 + 2^2 + \dots + k^2}$$

For the right hand side:

$$P(k) = \boxed{\frac{k(k+1)(2k+1)}{6}}$$

Solution: $P(k) = 1^2 + 2^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$

c.

For the left hand side:

$$\sum_{i=1}^{k+1} i^2 = 1^2 + 2^2 + \dots + k^2 + (k+1)^2$$

Let $\frac{k(k+1)(2k+1)}{6} = 1^2 + 2^2 + \dots + k^2$.

$$\sum_{i=1}^{k+1} i^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2$$

For the right hand side:

$$\frac{(k+1)((k+1)+1)(2(k+1)+1)}{6} = \boxed{\frac{(k+1)(k+2)(2k+3)}{6}}$$

Solution:

d.

In a proof by mathematical induction that the formula holds for every integer $n \geq 1$, we suppose that the formula is true for a particular but arbitrarily chosen integer k , and if true, it must also be true for $k+1$, for both the left-hand side and the right-hand side.

Solution: See above.